

Changes in soil properties and salt tolerance of safflower in response to biochar-based metal oxide nanocomposites of magnesium and manganese

Salar Farhangi-Abriz, Kazem Ghassemi-Golezani^{*,1}

Department of Plant Eco-physiology, Faculty of Agriculture, University of Tabriz, Tabriz, Iran

ARTICLE INFO

Edited by: Professor Bing Yan

Keywords:

Biochar
Magnesium oxide
Manganese oxide
Osmolytes
Plant growth
Sodium

ABSTRACT

This original research was performed to assess the possible effects of solid biochar (25 g biochar kg⁻¹ soil) and biochar-based nanocomposites (BNCs) of magnesium oxide (25 g BNC-MgO kg⁻¹ soil), manganese oxide (25 g BNC-MnO biochar kg⁻¹ soil) and combined use of these nanocomposites (12.5 g BNC-MgO + 12.5 g BNC-MnO kg⁻¹ soil) on soil properties and salinity (non-saline, 6 and 12 dSm⁻¹) tolerance of safflower plants (*Carthamus tinctorius* L.). Application of biochar, particularly BNCs increased the pH and cation exchange capacity of soil, and the contents of water, potassium, calcium, magnesium, manganese, chlorophyll (a & b), nutrients uptake, water use efficiency and plant growth. Sodium adsorption ratio (SAR), exchangeable sodium percentage (ESP) of soil, sodium absorption rate of plants and osmolyte production (soluble carbohydrates and proteins, proline and glycine betaine) under 6 and 12 dSm⁻¹ salinities were decreased by biochar and BNCs treatments. Sodium sorption capacity of BNCs was much higher than the solid biochar, which reflected the superiority of BNCs in decreasing sodium uptake by plants. The combined application of BNC-MgO + BNC-MnO proved to be the preferable treatment for decreasing salt toxicity in safflower. Biochar and BNCs improved root and shoot growth by lowering SAR, ESP, sodium absorption rate of plants and osmotic stress under saline conditions. These results conclude that BNCs can enrich the plant cells with nutrients, increase the nutrients absorption rate and maintain the plant tissue water content at an optimum level to improve plant growth under salt stress.

1. Introduction

Biochar is a carbon-rich and low-cost product obtained from the thermal degradation of natural resources in an oxygen limited condition (Ghassemi-Golezani and Farhangi-Abriz, 2019; Guo et al., 2020). Various carbon-based materials, such as plant litter, wood residues, animal manure, crop residues, and waste products can be used for biochar production. Interactions between biochar, soil biota, organic matter, and bioavailability of nutrients for plants are known to happen in a short period of time after addition to the soil (Wang et al., 2020). Biochar modifies soil pH and increases water holding capacity, cation exchange potential and microbial activity in the rhizosphere. These changes usually have beneficial effects on bioavailability of nutrients and crop performance, including crop health and yield (Ali et al., 2017). Application of biochar has a great potential to enhance plant tolerance of environmental stresses such as drought and salinity by changing soil characters, nutrient supply and lowering sodium uptake (Ali et al., 2017; Farhangi-Abriz and Torabian, 2017).

Salinity is a major abiotic stress that threatens plant production and portraits a serious challenge to food and feed security in the world. In general, salt stress restricts plant growth by enhancing sodium ion toxicity, nutrient imbalances, oxidative damages and osmotic stress (Abbas et al., 2018; Farhangi-Abriz et al., 2020). Addition of biochar to the soil is a practical way to enhance salt tolerance of plants through changes in soil properties such as decreased sodium adsorption ratio (SAR), exchangeable sodium percentage (ESP) and sodium uptake by plants, and also by enriching cells with different nutrients for plant growth (Amini et al., 2016; Ali et al., 2017). Ali et al. (2017) reported that biochar has a great ability to sorb sodium ions. This mechanism can be helpful in diminishing sodium uptake of plants under saline conditions. Addition of biochar to the soil significantly decreases sodium, but increases potassium content in plants (Ali et al., 2017). Farhangi-Abriz and Torabian (2017) showed that addition of biochar to the soil enhances water uptake by plants and reduces salinity-dependent osmotic stress in common bean plants. Successful absorption of nutrients is an important factor for rising growth and salt tolerance of plants

* Corresponding author.

E-mail address: golezani@gmail.com (K. Ghassemi-Golezani).

¹ <https://orcid.org/0000-0001-9560-1869>.

(Ghassemi-Golezani and Farhangi-Abri, 2018). Biochar could be also useful for enhancing bioavailability of some important nutrients such as magnesium and manganese in soil under normal and stressful conditions (Biederman and Harpole, 2013).

Magnesium is a part of chlorophyll structure and has an important role in safeguarding of ribosome molecular structure in plant cells and activation of enzymes. The sufficient level of magnesium is vital for salt tolerance of plants (Torabian et al., 2018; Javaid et al., 2019). There are many reports about the beneficial impacts of magnesium on rising salt tolerance of plants. Rivelli et al. (2010) reported that application of magnesium improved sunflower growth by increasing chlorophyll content and gas exchange capacity of leaves under salt stress. Application of magnesium silicate to the soil increased shoot and leaf growth of salt stressed castor oil plants (de Brito Neto et al., 2012). Manganese is another important nutrient in plant life cycle, which has some critical roles on rising salt tolerance of plants. Manganese treatment improved ion homeostasis, antioxidant defense and glyoxalase systems of rice seedlings in saline condition (Rahman et al., 2016). Shahi and Srivastava (2018) stated that treatment of mung bean plants by manganese ions significantly enhanced photosynthetic pigments, nitrate reductase activity and plant growth under salt stress. In a recent study, manganese application in nano form increased root growth and Mn-SOD gene expression in *Capsicum annuum* under salinity (Ye et al., 2020).

Production of biochar-based nanocomposites (BNCs) can create new complex materials, which combine the benefits of biochar with nanomaterials. Recent reports stated that physico-chemical modification processes such as depositing, adding or developing nutrients on biochar structure, could significantly enhance the impact of biochar on carbon sequestration, wastewater treatment and soil fertility (Tan et al., 2016). Previous studies showed that enriching biochar with different nutrients in both nano and/or normal forms can change biochar performance, nutrient bioavailability and plant growth. For example, nitrogen has been incorporated on biochar to produce biochar-based slow release nitrogen fertilizers with excellent water retention capacities (Wen et al., 2017). Enriching biochar with silicon mitigated the arsenic toxicity and improved the growth of spinach plants (Zama et al., 2018). Application of enriched biochars with magnetic iron nanoparticles significantly increased nutrient uptake and growth of wheat plants (Joseph et al., 2015). Biochar enrichment with phosphorus ions improved its performance and increased lettuce growth in alkaline soil (Gunes et al., 2014). Sulfur-enriched biochar increased the biochar nutrient releasing ability, soil fertility, plant nutrient content, and biomass in soybean and maize plants (Zhang et al., 2017).

Safflower (*Carthamus tinctorius* L.) is considered to be a moderately salt-tolerant plant, and has been cultivated in some parts of the central Asia and Iran, where salt toxicity inhibits growth of many other plants (Yeilaghi et al., 2012; Hussain et al., 2016). Seeds of this plant are a good source of high-quality edible oil. The productivity of safflower may be decreased under salt stress (Yeilaghi et al., 2012), but some agronomic practices such as changing soil properties and increasing soil fertility could be beneficial for improving salt tolerance and performance of this plant (Machado and Serralheiro, 2017). Based on successful roles of biochar in decreasing detrimental effects of salt stress on different plants and importance of magnesium and manganese in plant life cycle, this research was conducted for the first time to examine the possible impacts of enriching biochar with magnesium and manganese nano-oxides (biochar-based metal oxide nanocomposites) on safflower plant (*Carthamus tinctorius* L.) performance in non-saline and saline conditions.

2. Materials and methods

2.1. Production of biochar-based metal oxide nanocomposites

The biochar was purchased locally and biochar-based nanocomposites of magnesium and manganese oxides were produced according to the methods of Shen et al. (2019) and Song et al. (2014),

respectively. Initially biochar was separately mixed with magnesium chloride hexahydrate (0.01 M $\text{MgCl}_2 \cdot 6\text{H}_2\text{O}$) and potassium permanganate (0.01 M KMnO_4) at a ratio of 1 g:40 ml¹ and stirred for 4 h. The mixed materials were oven dried at 80 °C for 5 h and thereafter were heated at 460 °C. The mixtures were then washed thoroughly with distilled water to remove impurities, and then dried at 80 °C for 3 h. All of the chemical materials were purchased from Sigma-Aldrich Chemicals company (Missouri, USA).

The pH values of biochar and BNCs were determined by a pH meter (Model: HI 99121, Hanna Instrument, USA) in a 1:2.5 (w/v) ratio of char materials to deionized water. Total C, N, H and O were measured by a CHNOS elemental analyzer (Elementar-group, Hanau, Germany). The cation exchange capacity was assayed using the ammonium acetate method (Hendershot et al., 1993). Nutrient contents were determined by ICP Optical Emission Spectrometry (5800 ICP-OES Instrument, Hitachi, Tokyo, Japan) (Zhang et al., 2017). The Brunauer–Emmett–Teller (BET) surface area and pore size distributions were determined by the Quantachrome Autosorb Automated Gas Sorption System (White, 2010). The particle sizes of MgO and MnO within the biochar matrix were calculated by the Scherrer formula (Yao et al., 2013). Thermal stability of biochar and prepared metal oxide nanocomposites was determined by TGA/DSC analysis, and maximum weight loss (because of the CO_2 , CO and CH_4 emissions and decomposition of organic compounds) due to heat treatment was considered as maximum stability of these materials (Lateef et al., 2019). The sodium sorption capacity of biochar and BNCs was assayed by the method of Rostamian et al. (2015). A sample of 200 mg from each char material was added to 25 ml of 0.2 M NaCl solution, and then it was shaken for 12 h. The solution was filtered using Whatman 42 filter paper and the sodium concentration in the filtrate was measured using a Corning 410 flame photometer (Corning, New York, USA). The main physico-chemical properties of biochar and BNCs are presented in Table 1.

2.2. Experimental conditions

A greenhouse experiment was conducted with the day and night mean temperatures of 24 and 20 °C, respectively, 39% air humidity, 140 Wm^{-2} light intensity and about 13 h photoperiod. Three levels of sodium chloride salinities [non-saline (1.32 dSm⁻¹), 6 dSm⁻¹ (moderate salinity) and 12 dSm⁻¹ (severe salinity)] were adjusted on the basis of saturated soil paste extract. The levels of soil salinity were selected based on salt stress tolerance of safflower (Ghassemi-Golezani et al., 2020) and appropriate pretests. The five levels of biochar-based metal oxide nanocomposites [non-biochar, 25 g kg⁻¹ solid biochar, 25 g kg⁻¹ biochar-based nanocomposite of magnesium oxide (BNC-MgO), 25 g kg⁻¹ biochar-based nanocomposite of manganese oxide (BNC-MnO) and 12.5 g kg⁻¹ biochar-based nanocomposite of magnesium oxide + 12.5 g kg⁻¹ biochar-based nanocomposite of manganese oxide] were applied to the soil. The experimental soil (Table 1) was well mixed with BNCs and solid biochar according to the treatments and transferred to the polyethylene pots (25 × 25 cm with a capacity of 5.5 kg of soil). Then, about 10 seeds of safflower (*Carthamus tinctorius* L. cv Sina) were sown in each pot and irrigated with tap water to achieve 100% field capacity (FC). After seedling establishment, the plants were thinned and only 4 plants were kept in each pot. About 5 g of a fertilizer (Master 20–20–20-Valagro-Italy) was dissolved in a liter of tap-water (EC = 0.6 dSm⁻¹ and pH = 7.1) and added to the pots (250 ml per pot). At full flowering stage (75 days after the sowing), two plants from each pot were taken out and different biochemical and physiological measurements were performed.

2.3. Soil CEC, pH, SAR and ESP

Ammonium acetate method was used to assess the cation exchange capacity (CEC) of soil in each pot (Hendershot et al., 1993) and the soil pH was also measured by a pH meter (Model: HI 99121, Hanna

Table 1

Some physico-chemical properties of the experimental soil, biochar and biochar-based nanocomposites of magnesium (BNC-MgO) and manganese oxides (BNC-MnO).

Soil	Biochar	BNC-MgO	BNC-MnO
Texture	Silty loam	N (%)	0.52
pH	6.9	C (%)	40.25
EC (dSm ⁻¹)	1.32	H (%)	1.88
OC (g kg ⁻¹)	11.1	O (%)	28.95
Total N (%)	0.09	P (mg kg ⁻¹)	8120
P (mg kg ⁻¹)	31.2	S (mg kg ⁻¹)	5560
K (mg kg ⁻¹)	187.1	Na (mg kg ⁻¹)	1.80
Mg (mg kg ⁻¹)	70.4	K (mg kg ⁻¹)	3610
Mn (mg kg ⁻¹)	10.80	Ca (mg kg ⁻¹)	3640
CEC (cmol kg ⁻¹)	18.80	Mg (mg kg ⁻¹)	3430
Moisture (%)	1	Mn (mg kg ⁻¹)	29.64
		Cl (mg kg ⁻¹)	10.40
		CEC (cmol kg ⁻¹)	25.20
		pH	7.1
		S _{BET} (m ² g ⁻¹)	108.2
		V _{tot} (cm ³ g ⁻¹)	0.106
		AVP (nm)	3.81
		MgO size (nm)	23–60
		Stability (°C)	490
		Na-SC (mg g ⁻¹)	91.4
		Moisture (%)	<1

EC: electrical conductivity; OC: organic carbon; CEC: cation exchange capacity; NA: not available;

S_{BET}: BET surface area; V_{tot}: total pore volume; AVP: average pore size; Na-SC: sodium sorption capacity.

Instrument, USA). Soil nutrient contents were determined according to standard protocols of Jones (1999), and then SAR and ESP were calculated by the following formulas:

$$\text{SAR} = \text{Na} / \sqrt{\text{Ca} + \text{Mg}} / 2$$

$$\text{ESP} = \text{Na} / \text{CEC} \times 100$$

2.4. Water use efficiency, leaf area, chlorophyll content and plant growth

Plant water use up to flowering (75 days after the sowing) was estimated from the sum of the per diem water intake by the pots up to full flowering stage. Water use efficiency (WUE) was calculated as:

$$\text{WUE (g l}^{-1}\text{)} = (\text{Plant biomass at full flowering stage}) / \text{Total water used up}$$

Leaf area (LA) was measured by a portable area meter (model ADC-AM 300-United Kingdom) as cm² per plant. Chlorophyll content of leaf was assayed according to the method of Arnon (1949). Briefly 250 mg of each leaf sample was smashed and homogenized in 2 ml acetone (80%), and then centrifuged at 12,000 g for 20 min at 4 °C. The absorbance of upper liquid phases was recorded at 645 and 663 nm with UV-Visible Spectrophotometer (Dynamica, Halo DB-20 United Kingdom). Root length was measured by a ruler, then root and shoot dry weights were assayed after oven drying at 80 °C for 48 h.

2.5. Water content and osmolytes

Safflower roots and leaves were collected and weighed (FW), and then oven dried for 48 h at 80 °C and reweighed (DW). The roots and leaves water content (RWC and LWC) were determined by the following formula:

$$\text{RWC and LWC} = [(\text{FW} - \text{DW}) / \text{FW}] \times 100$$

Proline content of safflower root and leaf tissues was determined according to the method of Bates et al. (1973). Initially, 200 mg of root or leaf sample were homogenized in 4 ml of 3% sulphosalicylic acid and then 1.5 ml of the prepared samples were transferred into a falcon plastic tube and then 1.5 ml of ninhydrin and 1.5 ml of glacial acetic acid were added to this mixture. The mixture was heated at 100 °C for 1 h. Next, the mixture was cooled at 25 °C and the prepared sample was extracted with toluene, and the absorbance of upper liquid phase was

read at 520 nm with UV-Visible Spectrophotometer (Dynamica, Halo DB-20, United Kingdom). Glycine betaine content in roots and leaves was measured by the methods of Grieve and Grattan (1983). Briefly, 200 mg of root or leaf samples of safflower were smashed and mixed with 4 ml toluene-water mixture of 0.05% V/V in a 20 ml falcon plastic tube. Prepared samples were shaken for 12 h at 25 °C and then 0.1 ml of potassium tri-iodide and 1 ml of 2 N HCl were added to the mixtures and optical density of organic layer was read at 365 nm. The soluble carbohydrates and proteins in safflower roots and leaves were assayed by the method of Kochert et al. (1978) and Bradford (1976), respectively.

2.6. Nutrient content and absorption rate

Safflower roots and leaves were oven dried at 80 °C for 48 h and powdered. The powdered samples (200 mg from each sample) were burned for 5 h at 560 °C and the ashes were digested in 10 ml 1 M HCl at 25 °C for 24 h. Sodium, potassium, calcium, magnesium and manganese contents were measured by atomic absorption spectrophotometry (Shimadzu model: AA-7000, Kyoto, Japan). The digested plant samples were analyzed alongside with standard reference material of the National Institute of Standard and Technology. Nutrient absorption rate (AR) was calculated by the method of Silva et al. (2017) with some modifications, using the following formula:

$$\text{AR} = (\text{TNC at PHT}) - (\text{TNC at SET}) / \text{days from SET to PHT}$$

Where AR is absorption rate (μg day⁻¹), TNC is total nutrient content (μg g⁻¹ dry weight), PHT is plant harvesting time and SET is seedling establishment time.

2.7. Statistical analysis

The data were analyzed (Two-Way ANOVA – factorial arrangement on the basis of randomized complete block design with three replicates) by the MSTAT-C software (East Lansing- Michigan State University-USA) and means were compared by Duncan's Multiple Range test at $p \leq 0.05$. Principal component analyses (PCA-Pearson correlation) for soil parameters, safflower growth, nutrients and osmolytes of roots and leaves were performed, using the XLSTAT 2017 software (Addinsoft-New York-USA). This multivariate analysis reveals the association of different parameters with each other. The position of each parameter on PCA coordinates indicates the relative impact of this parameter on other

Table 2
The CEC (cmol kg⁻¹), pH, SAR and ESP (%) of soil and safflower root length (cm) root and shoot dry weights (g per plant), leaf area (cm² per plant) chlorophylls content (mg g⁻¹DW) and water use efficiency (g DW per liter) under different levels of salt stress and biochar-based nanocomposite treatments.

Salinity	Soil treatments	CEC	pH	SAR	ESP	Root length	Root DW	Shoot DW	Leaf area	Chl a	Chl b	WUE
Non-saline	Non-biochar	18.34 ± 0.33 ef	6.87 ± 0.03 cde	4.94 ± 0.17 e	6.67 ± 0.16 g	33.40 ± 0.30 e	4.16 ± 0.17 d	6.31 ± 0.15 d	325 ± 3.92 d	1.96 ± 0.03 d	1.11 ± 0.03 b	2.39 ± 0.07 d
	Biochar	19.00 ± 0.57 ef	6.90 ± 0.05 cde	3.13 ± 0.06 fg	5.24 ± 0.13 h	36.73 ± 0.64 c	5.01 ± 0.29 c	6.84 ± 0.16 BCE	344 ± 3.05 c	2.10 ± 0.06 c	1.12 ± 0.03 ab	2.67 ± 0.08 c
	BNC-MgO	21.00 ± 0.58 BCE	7.03 ± 0.03 ab	2.33 ± 0.06 h	4.60 ± 0.20 i	37.93 ± 0.58 b	6.00 ± 0.28 b	7.20 ± 0.17 ab	365 ± 2.88 b	2.26 ± 0.03 b	1.15 ± 0.02 ab	2.87 ± 0.10 b
	BNC-MnO	21.67 ± 0.34 abc	7.02 ± 0.04 ab	2.94 ± 0.08 g	4.66 ± 0.17 i	34.36 ± 0.32 d	5.84 ± 0.44 b	6.58 ± 0.25 cd	355 ± 2.96 BCE	2.23 ± 0.03 b	1.17 ± 0.03 ab	2.85 ± 0.09 BCE
6 (dSm ⁻¹)	BNC-MgO + BNC-MnO	22.67 ± 0.33 a	7.03 ± 0.03 ab	2.55 ± 0.05 h	4.67 ± 0.16 i	38.40 ± 0.31 a	6.75 ± 0.40 a	7.43 ± 0.23 a	384 ± 3.05 a	2.36 ± 0.03 a	1.19 ± 0.02 a	3.16 ± 0.07 a
	Non-biochar	17.65 ± 0.32 fg	6.61 ± 0.05 g	6.00 ± 0.10 c	11.45 ± 0.27 b	25.36 ± 0.31 k	3.00 ± 0.05 gh	4.26 ± 0.15 jk	282 ± 1.52 g	1.78 ± 0.01 fgh	0.93 ± 0.02 cd	1.69 ± 0.04 gh
	Biochar	18.35 ± 0.29 ef	6.80 ± 0.05 ef	3.44 ± 0.06 f	8.65 ± 0.16 e	29.40 ± 0.30 h	3.34 ± 0.09 fg	5.11 ± 0.07 fgh	294 ± 2.90 f	1.85 ± 0.02 ef	0.92 ± 0.02 cd	1.89 ± 0.06 f
	BNC-MgO	20.31 ± 0.34 cd	6.97 ± 0.03 abc	2.90 ± 0.03 g	7.52 ± 0.23 f	30.40 ± 0.25 g	3.62 ± 0.07 ef	5.56 ± 0.18 ef	313 ± 3.17 de	1.88 ± 0.03 de	0.95 ± 0.01 c	1.99 ± 0.02 ef
12 (dSm ⁻¹)	BNC-MnO	20.66 ± 0.31 bcd	7.06 ± 0.03 a	3.24 ± 0.07 fg	7.41 ± 0.20 f	28.40 ± 0.32 i	3.51 ± 0.06 efg	5.26 ± 0.16 fg	303 ± 1.85 de	1.89 ± 0.04 de	0.95 ± 0.01 c	2.00 ± 0.05 ef
	BNC-MgO + BNC-MnO	22.00 ± 0.58 ab	7.07 ± 0.03 a	2.94 ± 0.07 g	7.10 ± 0.10 f	32.36 ± 0.33 f	3.90 ± 0.07 de	5.83 ± 0.20 e	325 ± 2.88 d	1.91 ± 0.03 de	0.96 ± 0.02 c	2.16 ± 0.06 e
	Non-biochar	15.80 ± 0.46 h	6.45 ± 0.03 h	8.09 ± 0.05 a	12.86 ± 0.06 a	21.46 ± 0.29 i	2.03 ± 0.04 i	4.11 ± 0.11 k	230 ± 5.78 j	1.63 ± 0.03 i	0.87 ± 0.04 d	1.41 ± 0.09 i
	Biochar	16.77 ± 0.53 gh	6.71 ± 0.04 fg	6.66 ± 0.06 b	10.44 ± 0.26 c	25.40 ± 0.34 k	2.60 ± 0.10 h	4.57 ± 0.08 jk	251 ± 6.00 i	1.72 ± 0.02 h	0.86 ± 0.02 d	1.61 ± 0.08 h
F test for salinity treatments	BNC-MgO	18.60 ± 0.50 ef	6.84 ± 0.03 de	4.80 ± 0.10 c	9.41 ± 0.30 d	28.40 ± 0.38 j	3.02 ± 0.02 gh	4.71 ± 0.12 hij	270 ± 5.77 h	1.80 ± 0.01 fgh	0.90 ± 0.03 cd	1.68 ± 0.07 gh
	BNC-MnO	18.75 ± 0.37 ef	6.97 ± 0.03 abc	5.74 ± 0.14 c	9.55 ± 0.24 d	26.50 ± 0.33 j	2.75 ± 0.05 h	4.60 ± 0.05 ijk	260 ± 5.81 i	1.76 ± 0.03 gh	0.88 ± 0.02 cd	1.67 ± 0.08 gh
	BNC-MgO + BNC-MnO	19.50 ± 0.28 de	6.94 ± 0.02 bcd	5.32 ± 0.34 d	8.65 ± 0.34 e	29.50 ± 0.27 h	3.14 ± 0.09 fgh	5.00 ± 0.04 jhi	294 ± 2.90 f	1.82 ± 0.01 efg	0.91 ± 0.03 cd	1.82 ± 0.09 fg
	F test for soil treatments	51.47**	39.97**	806.36**	1453.54**	9861.77**	365.99**	288.27**	668.02**	347.95**	531.93**	431.44**
F test for salinity × soil treatments		47.79**	63.73**	277.04**	274.21**	1489.24**	34.46**	25.12**	78.91**	34.60**	8.81**	29.75**
		2.45*	4.75**	5.79**	11.82**	57.62**	4.51**	4.35**	3.67**	4.42**	4.93**	5.24**

Data represents the average of three replicates (n = 3) ± standard error. Different letters indicate significant differences by Duncan's Multiple Range test at p ≤ 0.05.

BNC-MgO: biochar-based nanocomposite of magnesium oxide; BNC-MnO: biochar-based nanocomposite of manganese oxide; DW: dry weight.

* significant at p ≤ 0.05, respectively.

** significant at p ≤ 0.01, respectively.

parameters. All figures were drawn by the Excel 2019 (Microsoft Corporation-USA).

3. Results

3.1. Biochar and biochar-based nanocomposites properties

The contents of magnesium and manganese in biochar structure were increased, but potassium, phosphorous and calcium were slightly decreased in composition processes (Table 1). This composition improved the CEC, pH, Brunauer–Emmett–Teller (BET) surface area, total pore volume and average pore size of biochar. The MgO and MnO sizes in biochar matrix were 23–60 and 35–64 nm, respectively. Enriching biochar with magnesium and manganese slightly decreased the thermal stability of the resultant nanocomposites. The sodium sorption capacity of biochar was noticeably enhanced in the composition process. The highest sodium sorption capacity, surface area and CEC were observed in biochar-based nanocomposite of magnesium oxide.

3.2. Soil properties

Enhancing salt toxicity of soil caused a decline in the CEC and pH of soil, but SAR and ESP were increased by rising salt level (Table 2). Treatment of soil with biochar and BNCs increased the soil CEC and pH under both saline and non-saline conditions. The combination of biochar with metal oxide nanocomposites showed the highest CEC in experimental soil. All the metal oxide nanocomposite treatments showed a similar effect on rising soil pH under non-saline condition and 6 dSm⁻¹ salinity, however BNC-MgO had comparatively better effect than other treatments on rising soil pH under 12 dSm⁻¹. Addition of biochar and biochar-based nanocomposites of magnesium and manganese oxides dramatically decreased the SAR and ESP in experimental soil. The BNC-MgO and BNC-MgO + BNC-MnO showed the lowest level of SAR in experimental soil. Combined use of biochar-based metal oxide nanocomposites was the superior treatment on decreasing ESP of soil, especially under 12 dSm⁻¹.

3.3. Plant performance and water use efficiency

The root length, root and shoot dry weights, leaf area, contents of chlorophyll a and b and water use efficiency were decreased as a result of salt increment (Table 2). Application of solid biochar and particularly BNCs improved the root and shoot growth, leaf area, contents of chlorophyll a and b and water use efficiency in safflower plants. Combined use of metal oxide nanocomposites had the highest impact on rising safflower growth under saline and non-saline conditions. This treatment improved the root, shoot and leaf growth under severe salinity by about 54%, 21% and 27%, respectively compared to the non-biochar treatment. The BNC-MgO + BNC-MnO treatment was led to the highest level of water use efficiency by safflower plants, compared to different non-biochar and other BNC treatments.

3.4. Water content and osmolytes

Rising salinity increased the soluble proteins and carbohydrates, glycine betaine and proline contents, but decreased the water content in safflower leaves and roots (Table 3). Under saline conditions, application of biochar and BNCs both in individual and combined forms decreased the soluble proteins and carbohydrates, glycine betaine and proline content, but augmented the water content in safflower roots and leaves. Treating soil with biochar and BNCs enhanced osmolytes under non-saline condition. Addition of biochar and BNCs to the soil, did not produce a tangible effect on water content of safflower roots and leaves in unstressed plants. In most cases, there were no significant differences between the solid biochar and nanocomposite treatments on rising

Table 3

Means of osmolytes and water content of safflower plants under different levels of salt stress and biochar-based nanocomposites of magnesium and manganese oxides applications.

Salinity	Soil treatments	Soluble proteins (mg g ⁻¹ DW)		Soluble carbohydrates (mg g ⁻¹ DW)		Glycine betaine (mg g ⁻¹ DW)		Proline (mg g ⁻¹ FW)		Water content (%)	
		Root	Leaf	Root	Leaf	Root	Leaf	Root	Leaf	Root	Leaf
Non-saline	Non-biochar	4.03 ± 0.57 g	5.07 ± 0.59 g	4.00 ± 0.51 e	6.28 ± 0.88 g	1.32 ± 0.33 d	2.37 ± 0.34 g	11.00 ± 1.15 i	13.68 ± 1.45 g	88.59 ± 1.20 a	87.33 ± 0.88 a
	Biochar	6.00 ± 0.59 f	8.00 ± 0.61 f	6.64 ± 0.88 d	7.34 ± 0.85 g	2.00 ± 0.29 d	2.31 ± 0.31 g	13.66 ± 0.89 hi	15.58 ± 1.45 g	89.35 ± 0.89 a	88.00 ± 0.57 a
	BNC-MgO	5.94 ± 0.54 f	7.95 ± 0.34 f	5.66 ± 0.78 de	7.03 ± 0.58 g	1.37 ± 0.32 d	2.69 ± 0.37 g	14.65 ± 0.33 h	17.00 ± 1.15 fg	88.31 ± 0.88 a	87.10 ± 0.55 ab
	BNC-MnO	7.05 ± 0.58 ef	8.09 ± 0.57 f	6.58 ± 0.89 d	7.00 ± 0.44 g	2.29 ± 0.30 d	2.65 ± 0.33 g	14.31 ± 0.94 h	17.00 ± 2.08 fg	88.45 ± 0.87 a	86.00 ± 0.61 BCE
6 (dSm ⁻¹)	BNC-MgO + BNC-MnO	6.94 ± 0.54 ef	7.91 ± 0.54 f	6.49 ± 0.91 d	7.39 ± 0.54 g	1.67 ± 0.39 d	2.41 ± 0.28 g	15.00 ± 0.89 h	17.35 ± 1.40 fg	88.46 ± 0.85 a	87.67 ± 0.45 a
	Non-biochar	12.02 ± 0.47c	14.69 ± 1.20 BCE	15.29 ± 0.67 a	18.37 ± 1.45 a	8.00 ± 0.57 a	8.38 ± 0.67 b	31.00 ± 1.15 d	32.33 ± 1.34 d	82.00 ± 0.57 b	80.32 ± 0.91 d
	Biochar	7.64 ± 0.33 e	10.04 ± 0.55 e	9.31 ± 0.69c	12.36 ± 1.34 f	4.05 ± 0.56c	5.00 ± 0.54 f	18.35 ± 0.67 g	21.38 ± 1.26 ef	82.65 ± 0.67 b	83.35 ± 0.55c
	BNC-MgO	8.59 ± 0.36 d	12.01 ± 0.58 d	11.38 ± 0.59 BCE	14.65 ± 1.20 cde	5.35 ± 0.44 BCE	5.65 ± 0.57 def	22.00 ± 0.59 ef	25.29 ± 2.11 e	83.66 ± 0.66 b	84.40 ± 0.61 BCE
12 (dSm ⁻¹)	BNC-MnO	9.07 ± 0.44 d	12.36 ± 0.55 d	10.21 ± 0.58 BCE	14.31 ± 1.46 de	5.65 ± 0.52 b	5.29 ± 0.60 de	20.24 ± 0.65 fg	21.30 ± 1.42 ef	83.06 ± 0.68 b	83.67 ± 1.20c
	BNC-MgO + BNC-MnO	9.65 ± 0.37 d	12.98 ± 0.48 cd	11.40 ± 0.54 BCE	16.07 ± 1.00 b	5.28 ± 0.51 BCE	5.59 ± 0.24 def	23.68 ± 0.88 e	25.00 ± 1.18 e	83.57 ± 0.81 b	84.25 ± 0.90 BCE
	Non-biochar	17.03 ± 0.58 a	19.88 ± 0.55 a	16.62 ± 0.66 a	19.39 ± 0.91 a	9.00 ± 0.60 a	10.03 ± 0.58 a	57.65 ± 1.77 a	64.00 ± 2.31 a	73.32 ± 0.35 d	71.00 ± 0.58 g
	Biochar	12.31 ± 0.44c	12.89 ± 0.47 cd	10.41 ± 0.88 BCE	13.68 ± 0.84 e	4.61 ± 0.59 BCE	6.08 ± 0.54 def	36.00 ± 1.76c	36.68 ± 1.86c	77.67 ± 0.32c	74.67 ± 0.75 f
	BNC-MgO	13.96 ± 0.39 b	15.00 ± 0.61 BCE	11.64 ± 0.34 BCE	15.55 ± 0.94 BCE	5.71 ± 0.54 b	6.97 ± 0.54 cd	39.05 ± 1.00 BCE	43.66 ± 1.44 b	76.29 ± 0.67c	76.66 ± 0.58 ef
	BNC-MnO	14.02 ± 0.48 b	15.67 ± 0.48 b	10.68 ± 0.89 bc	15.32 ± 0.89 BCEd	5.28 ± 0.52 BCE	6.62 ± 0.39 cde	37.56 ± 0.34 BCE	39.00 ± 1.32c	76.65 ± 0.54c	77.34 ± 0.51 ef
	BNC-MgO + BNC-MnO	14.06 ± 0.51 b	16.06 ± 0.44 b	12.05 ± 0.22 b	15.69 ± 1.20 BCE	5.34 ± 0.59 BCE	7.63 ± 0.41 BCE	40.31 ± 0.88 b	44.06 ± 1.49 b	76.34 ± 0.78c	78.00 ± 1.52 de
	<i>F</i> test for salinity treatments	748.35**	210.06**	119.00**	868.00**	123.18**	183.06**	1039.97**	550.03**	451.21**	207.85**
	<i>F</i> test for soil treatments	19.26**	7.63**	9.30**	36.10**	12.42**	14.41**	49.93**	32.48**	3.55*	7.36**
	<i>F</i> test for salinity × soil treatments	20.66**	8.95**	8.31**	15.30**	5.25**	4.37**	26.71**	17.70**	2.61*	3.01*

Data represents the average of three replicates (n = 3) ± standard error. Different letters indicate significant differences by Duncan's Multiple Range test at p ≤ 0.05.

DW: dry weight; FW: fresh weight; BNC-MgO: biochar-based nanocomposite of magnesium oxide; BNC-MnO: biochar-based nanocomposite of manganese oxide.

* significant at p ≤ 0.05, respectively.

** significant at p ≤ 0.01, respectively.

Table 4
Means of sodium, potassium and calcium contents (mg g^{-1} DW), and their absorption rate ($\mu\text{g day}^{-1}$) in safflower plants under different levels of salt stress and biochar-based nanocomposites of magnesium and manganese oxides applications.

Salinity	Soil treatments	Sodium		Potassium		Calcium	
		Root	Leaf	Root	Leaf	Root	Leaf
Non-saline	Non-biochar	7.31 ± 0.88 g	5.29 ± 0.89 e	33.37 ± 0.92 c	35.68 ± 1.21 b	10.35 ± 90 d	8.41 ± 0.84 def
	Biochar	8.28 ± 0.91 g	5.58 ± 0.77 e	38.00 ± 1.73 b	38.61 ± 1.25 a	12.02 ± 0.58 BCE	10.28 ± 0.61 abc
	BNC-MgO	7.40 ± 0.84 g	5.31 ± 0.82 e	41.65 ± 1.20 a	41.58 ± 1.36 a	13.37 ± 0.77 a	11.30 ± 0.48 ab
	BNC-MnO	8.36 ± 1.05 g	5.69 ± 0.65 e	38.70 ± 1.20 b	40.71 ± 1.26 a	12.58 ± 0.42 ab	11.35 ± 0.57 ab
6 (dSm ⁻¹)	BNC-MgO + BNC-MnO	8.00 ± 1.15 g	5.65 ± 0.1101 e	40.30 ± 1.32 ab	41.02 ± 0.98 a	13.32 ± 0.53 a	11.72 ± 0.59 a
	Non-biochar	26.68 ± 0.67 d	24.07 ± 1.00 c	25.02 ± 0.58 fg	25.64 ± 0.89 ef	8.29 ± 0.89 e	6.70 ± 0.34 g
	Biochar	22.67 ± 0.75 e	19.02 ± 1.00 d	29.00 ± 0.64 de	31.00 ± 0.58 cd	11.28 ± 0.92 c	9.41 ± 0.53 cd
	BNC-MgO	20.70 ± 0.65 ef	16.61 ± 0.93 d	32.67 ± 0.88 c	34.05 ± 0.74 BCE	12.39 ± 0.89 abc	10.04 ± 0.58 BCE
12 (dSm ⁻¹)	BNC-MnO	19.61 ± 0.72 f	17.23 ± 0.78 d	31.62 ± 0.74 cd	31.32 ± 0.84 cd	11.64 ± 0.89 BCE	9.06 ± 0.61 cde
	BNC-MgO + BNC-MnO	20.31 ± 0.74 f	16.55 ± 0.44 d	33.00 ± 0.84 c	31.29 ± 0.99 cd	12.60 ± 0.57 ab	11.00 ± 0.52 ab
	Non-biochar	37.35 ± 0.64 a	33.07 ± 0.59 a	20.06 ± 0.34 h	20.70 ± 0.63 g	5.03 ± 0.58 g	4.68 ± 0.22 h
	Biochar	31.29 ± 0.18 b	27.00 ± 0.55 b	24.00 ± 0.74 g	25.03 ± 0.82 f	7.00 ± 0.71 f	6.70 ± 0.45 g
F test for salinity treatments	BNC-MgO	29.05 ± 0.43 c	24.36 ± 0.82 c	26.58 ± 0.59 efg	28.00 ± 1.04 ef	8.06 ± 0.35 e	7.63 ± 0.57 efg
	BNC-MnO	28.89 ± 0.69 c	24.23 ± 0.47 c	25.35 ± 0.93 fg	26.06 ± 0.64 ef	8.30 ± 0.44 e	7.03 ± 0.57 fg
	BNC-MgO + BNC-MnO	29.00 ± 0.56 c	22.41 ± 0.66 c	27.61 ± 0.58 ef	28.34 ± 0.86 de	10.08 ± 0.59 d	7.28 ± 0.48 fg
	F test for salinity treatments	1192.46**	805.09**	270.16**	271.06**	266.52**	97.31**
F test for soil treatments	Non-biochar	22.27**	25.14**	33.67**	25.62**	65.93**	26.89**
	Biochar	7.38**	7.34**	4.57**	2.66**	2.29*	2.45*

Data represents the average of three replicates ($n = 3$) ± standard error. Different letters indicate significant differences by Duncan's Multiple Range test at $p \leq 0.05$.

BNC-MgO: biochar-based nanocomposite of magnesium oxide; BNC-MnO: biochar-based nanocomposite of manganese oxide; AR: absorption rate.

* significant at $p \leq 0.05$, respectively.

** significant at $p \leq 0.01$, respectively.

water content of roots and leaves in safflower plants under 6 and 12 dSm⁻¹ salinities. Under high level of salinity, leaf water content of safflower was improved by about 3.6%, 5.6%, 6.3% and 7% in response to applied solid biochar, BNC-MgO, BNC-MnO and BNC-MgO + BNC-MnO treatments, respectively.

3.5. Nutrient contents

The sodium content in root and leaf tissues was increased, but potassium, calcium and magnesium contents were decreased by rising salt stress (Table 4, Fig. 1). Increasing salinity of soil up to 6 dSm⁻¹ did not significantly change the manganese content in safflower root and leaf, but it was decreased considerably by further increment of salt stress up to 12 dSm⁻¹. The sodium content of root and leaf in safflower plant did not affect by solid biochar and BNCs under non-saline condition, but these treatments significantly decreased sodium contents under saline conditions. The BNCs in comparison with biochar alone had a better effect on reducing sodium content in root and leaf tissues at the highest level of salinity.

Addition of solid biochar and BNCs significantly increased the potassium, calcium, magnesium and manganese contents in safflower roots and leaves (Table 4, Fig. 1) both in saline and non-saline soil conditions. In most cases, the highest potassium content was accumulated in plant tissues by applied BNC-MgO and combined use of nanocomposites under saline and non-saline conditions. Combined application of BNC-MgO and BNC-MnO increased the potassium content of roots and leaves under severe salinity by about 37% over non-biochar treatment. Soil treatment by BNCs showed a better effect than solid biochar on increasing calcium content in plant tissues, however in some cases these advantages were not significant. In general, combined form of nanocomposites was the superior treatment on improving calcium content of plants under saline and non-saline conditions. The BNC-MgO + BNC-MnO treatment enhanced the calcium contents of roots and leaves under high salinity by about 101% and 56%, respectively (Table 4).

The highest content of magnesium in plant tissues was achieved by applied BNC-MgO under all levels of salinity. In most cases, combined use of metal oxide nanocomposites ranked second in rising magnesium content of plant root and leaf, which was followed by solid biochar and manganese nanocomposite. The highest content of manganese in root and leaf tissues was recorded for BNC-MnO and BNC-MgO + BNC-MnO treatments. The BNC-MgO did not have significant ($p \leq 0.05$) impact on manganese content of safflower plants, compared to the solid biochar treatment. The BNC-MgO + BNC-MnO in comparison with non-biochar treatment enriched the safflower roots and leaves by accumulating about 123% and 72% more manganese under high level of salinity, and 101% and 91% under non-saline condition, respectively (Fig. 1c and d).

3.6. Nutrient absorption rate

Rising salinity enhanced the sodium absorption rate, but decreased the absorption rates of potassium, calcium, magnesium and manganese in safflower plants (Table 4, Fig. 1e and f). Soil treatment with biochar BNCs noticeably decreased the sodium absorption rate under 6 and 12 dSm⁻¹ salinities, but did not change this rate under non-saline condition. In fact, biochar and BNCs treatments significantly improved nutrients uptake (except sodium) over non-biochar treatment. In general, metal oxide nanocomposites showed better effects than solid biochar by diminishing sodium absorption rate under salt stress. Application of metal oxide nanocomposites in combined form lowered the sodium absorption rate, over non-biochar treatment, by about 28% and 27% under 6 and 12 dSm⁻¹ salinities, respectively. There were no significant differences between the BNCs decreasing with regard to sodium absorption rate under saline and non-saline conditions.

Potassium absorption rate was increased by biochar and BNCs under different levels of salinity and non-saline condition. These increments of potassium absorption rate were greater under high level of salinity. All

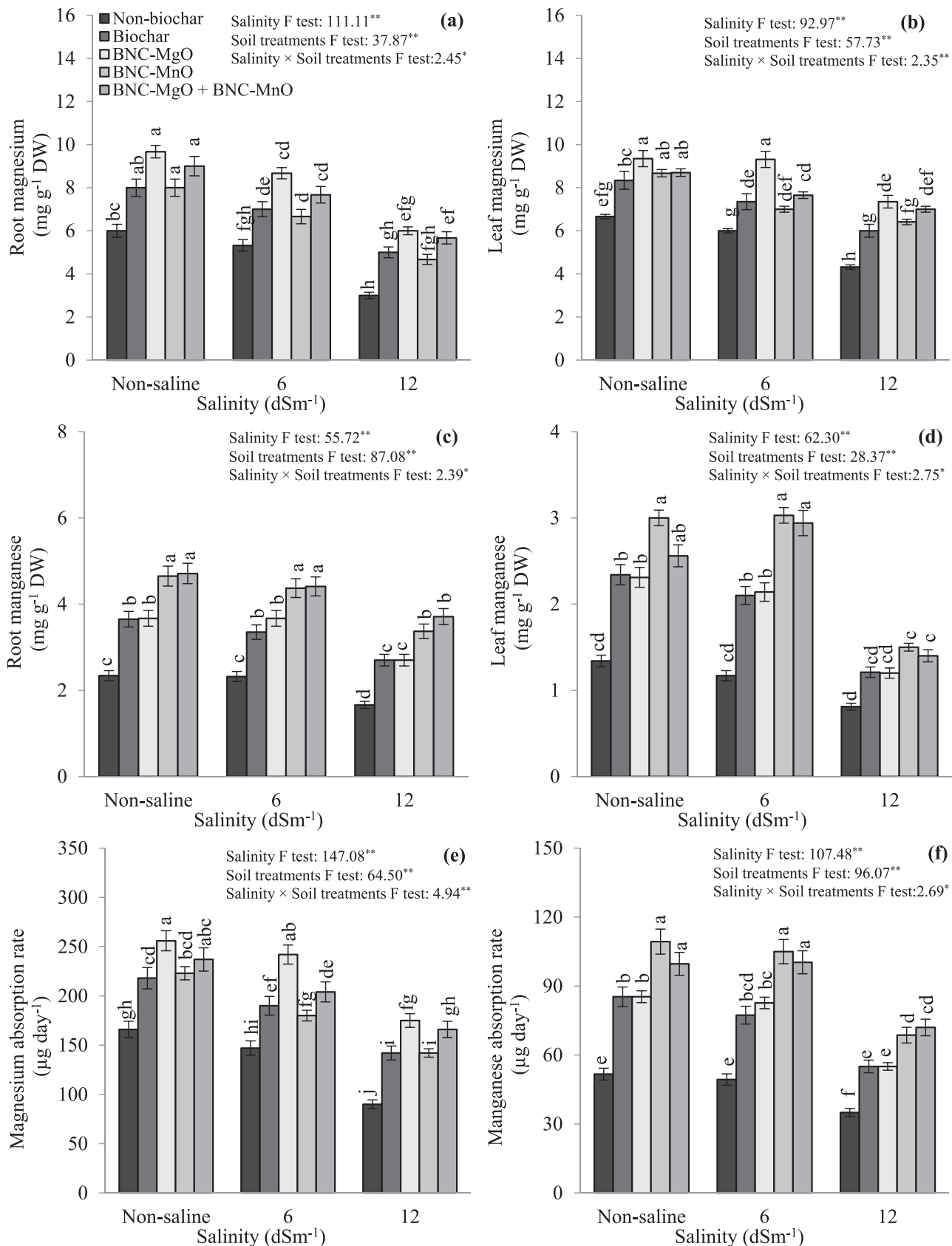


Fig. 1. Means of root magnesium (a), leaf magnesium (b), root manganese (c) and leaf manganese (d) contents and magnesium and manganese absorption rates (e and f) in safflower plants under different levels of salt stress and biochar-based nanocomposites of magnesium and manganese oxides applications. Data represents the average of three replicates ($n = 3$) $\pm$ standard error. Different letters indicate significant differences by Duncan multiple range test at $p \leq 0.05$. *, ** significant at $p \leq 0.05$ and $p \leq 0.01$, respectively. BNC-MgO: Biochar-based nanocomposite of magnesium oxide; BNC-MnO: biochar-based nanocomposite of manganese oxide DW: dry weight.

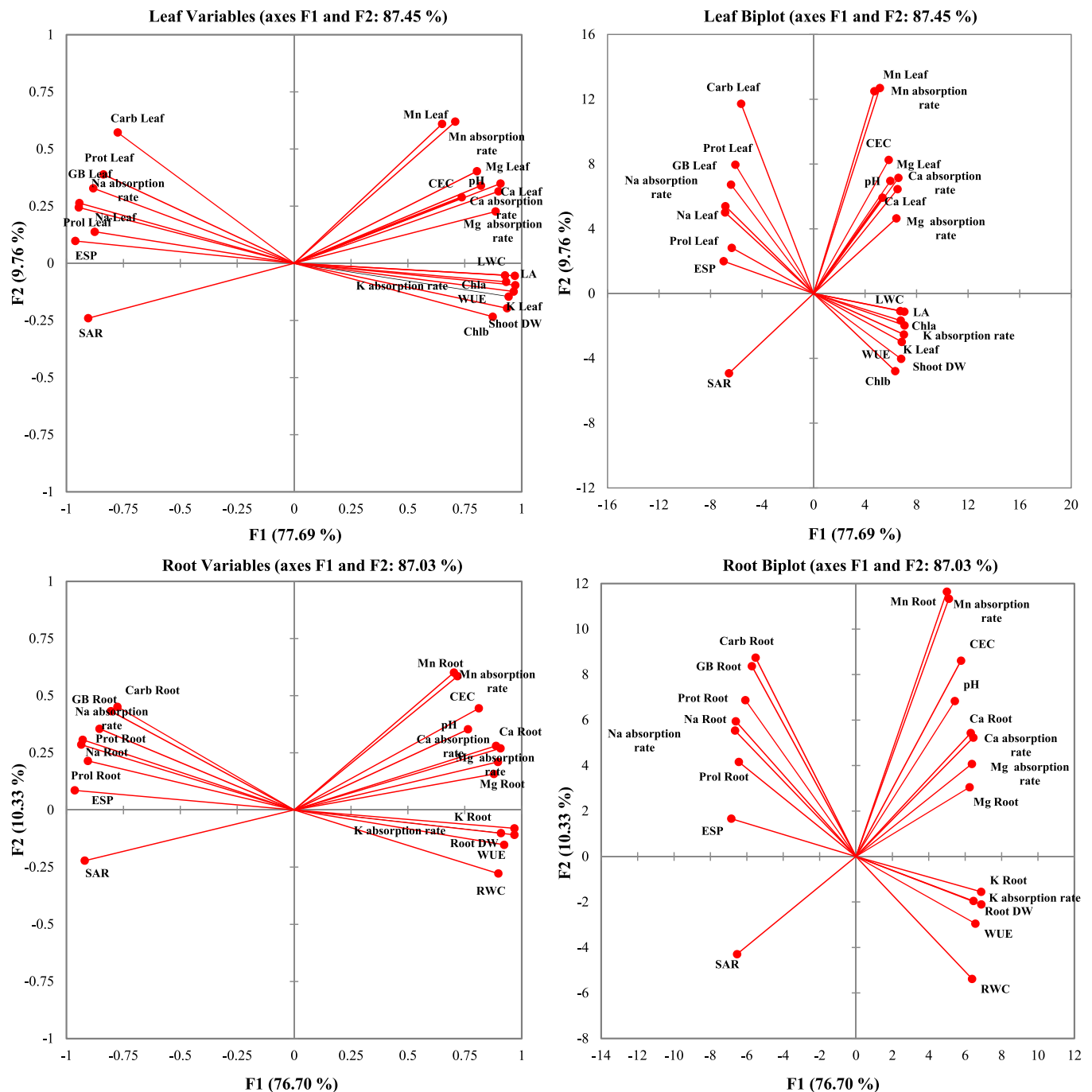


Fig. 2. Principal component analysis (PCA) of the leaf and root parameters of safflower plants. Carb: soluble carbohydrates; GB: glycine betaine; Prot: soluble proteins; Prol: proline; CEC: cation exchange capacity; ESP: exchangeable sodium percentage of soil; SAR: sodium adsorption ratio of soil; RWC: root water content; WUE: water use efficiency; DW: dry weight; Na: sodium; K: potassium; Ca: calcium; Mg: magnesium; Mn: manganese; Chl a: Chlorophyll a; Chl b: Chlorophyll b; LWC: leaf water content; LA: leaf area; DW: dry weight.

of the metal oxide nanocomposites had a similar effect on improving potassium absorption rate by safflower under different levels of salt stress. Potassium absorption rate was improved by about 23%, 39%, 29% and 43% under high level of salinity, and by about 11%, 22%, 15% and 19% under non-saline condition with solid biochar, BNC-MgO, BNC-MnO and the combined form of nanocomposites, respectively. Application of biochar and BNCs enhanced the calcium absorption rate in non-stressed and salt stressed plants, especially in severely salt stressed plants. Soil treatment with combined form of nanocomposites was the superior treatment for improving calcium absorption rate under salt stress conditions (Table 4). Magnesium and manganese absorption rates

were improved by solid biochar and BNCs treatments (Fig. 1e and f). The highest absorption rate of magnesium in safflower plants was achieved by BNC-MgO under different saline and non-saline conditions. The increment of magnesium absorption rate by solid biochar and nanocomposites was more evident under high level of salt stress. Soil treatments with BNC-MnO and combined form of nanocomposites resulted in the highest absorption rate of manganese under different levels of salt stress.

3.7. Principal component analysis (PCA)

The PCA was used to evaluate the relations between soil characters, biochemical and physiological parameters of safflower plants. The PCAs contributed up to 87% of variance in leaf parameters (Fig. 2). The PCA results revealed that sodium content of root and leaf, sodium absorption rate, osmolytes, ESP and SAR were negatively related, but potassium, calcium, magnesium, manganese, chlorophyll and water contents, and LA were positively correlated with plant growth. Safflower shoot and root growth had a strong positive correlation with water content, potassium absorption rate and chlorophyll content. Additionally, chlorophyll content was highly correlated with potassium content. Manganese, calcium and magnesium absorption rates had a strong correlation with CEC of soil, while the absorption rate of potassium had a positive correlation with water content in plant tissues. The ESP had the greatest impact on rising sodium absorption rate in safflower plants. Soluble proteins and proline content were more effective than soluble carbohydrates for improving osmotic adjustment of plants. A strong relation between osmolytes production and sodium content was found. Leaf area was highly correlated with water content and potassium uptake.

4. Discussion

Salinity by causing osmotic stress and nutrient imbalance decreased safflower growth (Table 2). High content of sodium ions in the plant tissues disorders osmotic regulation (Table 3) and nutrient balance (Table 4, Fig. 1), and imposes specific toxicity of sodium. The decrement of potassium content in plant tissues in response to excessive sodium ions is a well-known competitive process in plant roots, which is resulted from low uptake and efflux of potassium in root cells. Sodium-induced potassium efflux from mesophyll is eased by depolarization-activated outward rectifying potassium channel (Ghassemi-Golezani and Farhangi-Abri, 2018). The high levels of sodium ions in rhizosphere not only interfere with potassium uptake by the plant root, but also may disorder the integrity of root membranes and alter their selectivity. Reduction of plant growth in response to specific ion toxicity of sodium and imbalances of nutrients have been supported by a previous report on safflower (Hussain et al., 2016).

Biochar and metal oxide nanocomposite treatments improved the root and shoot growth of safflower plants (Table 2) by diminishing sodium absorption rate and its accumulation in plant cells (Table 4), and also by mitigating osmotic stress (Table 3) under salt stress. Osmotic stress is one of the main harsh impacts of salinity that decreases osmotic potential of plants. Plants have some especial mechanisms such as elevating osmolytes production to keep water potential at an optimum level for cellular activities under salt stress. In fact, osmolytes are compatible osmoprotectant solutes that increase the cell potential to preserve water without hindering the normal metabolism (Sharma et al., 2019). In this study, the content of osmolytes such as soluble proteins and carbohydrates, proline and glycine betaine were enhanced by increasing sodium content of plant tissues. The increment of osmolytes by rising sodium ions in plant tissues is a common response of plants under salt stress, reflecting the stimulation of osmolytes production by high sodium content of plants (Fig. 2). The diminishing sodium absorption rate led to a reduction in osmolytes in comparison with non-biochar treatments (Table 3). Moreover, biochar and especially its metal oxide nanocomposite forms have a great amount of total pore volume in their structure, which can help these materials to preserve water (Table 1). Reduction of sodium content in safflower plants in response to the biochar and nanocomposite treatments are the result of decreasing SAR and ESP of soil and sodium absorption rate under salt stress (Tables 2 and 4). BNCs had an excellent sodium sorption capacity and cation exchange potential, which can reduce ESP of soil (Tables 1 and 2). The increment of sodium sorption capacity of biochar in the composition process could be related to the enhancement of specific surface area and pore volume (Table 1). The specific surface area and

pore volume of biochar are important factors in controlling sodium sorption capacity of biochar (Rostamian et al., 2015).

Biochar and particularly BNCs have a great CEC which can increase the soil CEC and nutrient availability under salt stress. BNCs can improve the soil CEC with two well-known mechanisms: 1) these nanomaterials can improve the CEC of soil by increasing soil pH, and 2) high specific surface area of these materials can enhance the active surface area to exchange the nutrients in rhizosphere (Table 1). Agegnehu et al. (2016) reported that application of biochar noticeably enhanced the exchangeable cations and CEC of soil in a maize farmland. These researchers indicated that increasing soil CEC in response to the biochar applications could be related to adequate level of water in soil. Chintala et al. (2014) showed that biochar application increased soil pH and consequently improved soil CEC of soil. In most cases, soil CEC and pH have a positive correlation with each other (Ghassemi-Golezani and Farhangi-Abri, 2019).

Adding biochar and BNCs to the soil enriched the plant cells with different nutrients such as potassium, calcium, magnesium and manganese under salt stress (Table 4, Fig. 1). These improvements of nutrient content are the result of CEC enhancement, nutrient bioavailability and nutrient absorption rate in salt stressed plants (Table 4, Figs. 1 and 2). Accelerating nutrients absorption by BNCs are related with high specific surface area of these materials (Table 1). The magnesium, manganese and calcium contents of plants were strongly and positively correlated with CEC of the soil, while the potassium absorption rate was related with water availability (Fig. 2). Increasing soil pH by application of biochar and BNCs could be helpful in rising nutrient bioavailability and absorption rate by plants (Silber et al., 2010). Moreover, noticeable enhancements of magnesium and manganese contents of plants in response to the treatment of BNCs were likely related to the sizes of MgO and MnO in biochar matrix. The absorption of nano-sized nutrients is usually much easier than normal sized ones in rhizosphere (Siddiqui et al., 2015).

Increasing water use efficiency of plants in response to biochar and BNC treatments was related with the better growth of roots and shoots, resulting in the optimal use of water for growth. Improving chlorophyll content of salt stressed plants in response to biochar and BNC treatments was associated with high amount of magnesium and manganese in leaf tissues (Fig. 1b and d). Magnesium and manganese are very critical nutrients for biosynthesis of chlorophyll and successful growth of plants under non-stressful and stressful conditions (Farhangi-Abri and Ghassemi-Golezani, 2018). Tränkner and Jaghdani (2019) showed that adequate levels of magnesium ions had a significant effect on rising chlorophyll content and photosynthetic activities in wheat and sunflower seedlings. Similar report is available for feverfew plants (Farzadfar et al., 2017). The BNC-MgO and BNC-MgO + BNC-MnO were the superior treatments for improving some important soil characters such as soil pH and CEC under salt stress, helping plants to increase salt tolerance, nutrient uptake and growth (Tables 2 and 3, Fig. 1). Most of the beneficial effects of the BNCs were related to the reduction of ESP and sodium absorption rate of plants under salt stress, especially under severe salinity (Tables 2 and 4).

5. Conclusions

Soil treatment with solid biochar and particularly biochar-based metal oxide nanocomposites reduced the SAR, ESP and sodium absorption rate of plants under salt stress. These soil amendments increased the CEC of the soil and enriched plant cells with different nutrients such as potassium, calcium, magnesium and manganese and reduced the osmotic stress caused by salt stress. The solid biochar and metal oxide nanocomposite treatments showed a helpful effect on improving plant performance and water use efficiency under salt stress. The combined form of biochar-based nanocomposites was the superior treatment for improving root and shoot growth, salt tolerance and water use efficiency of the plants. Decrement of ESP and sodium absorption

rate of plants and increment of magnesium, manganese and potassium absorption rates and contents in plant tissues were the major advantages of metal oxide nanocomposites applications under salt stress. The results of this research clearly indicated that the biochar-based metal oxide nanocomposites with particular physicochemical properties such as high-level of CEC, sodium adsorption capacity and specific surface area had a great potential to enhance the stress tolerance of plants under salinity.

CRedit authorship contribution statement

Kazem Ghassemi-Golezani: Supervision, Experimental design, Writing. **Salar Farhangi-Abri:** Experimental work, Data analyzing, Writing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

We appreciate the University of Tabriz for supporting this research.

References

- Abbas, T., Rizwan, M., Ali, S., Adrees, M., Zia-ur-Rehman, M., Qayyum, M.F., Ok, Y.S., Murtaza, G., 2018. Effect of biochar on alleviation of cadmium toxicity in wheat (*Triticum aestivum* L.) grown on Cd-contaminated saline soil. *Environ. Sci. Pollut. Res.* 25, 25668–25680.
- Aegenehu, G., Bass, A.M., Nelson, P.N., Bird, M.I., 2016. Benefits of biochar, compost and biochar-compost for soil quality, maize yield and greenhouse gas emissions in a tropical agricultural soil. *Sci. Total Environ.* 543, 295–306.
- Ali, S., Rizwan, M., Qayyum, M.F., Ok, Y.S., Ibrahim, M., Riaz, M., Arif, M.S., Hafeez, F., Al-Wabel, M.I., Shahzad, A.N., 2017. Biochar soil amendment on alleviation of drought and salt stress in plants: a critical review. *Environ. Sci. Pollut. Res. Int.* 24, 12700–12712.
- Amini, S., Ghadiri, H., Chen, C., Marschner, P., 2016. Salt-affected soils, reclamation, carbon dynamics, and biochar: a review. *J. Soil Sediment* 16, 939–953.
- Arnon, D.I., 1949. Copper enzymes in isolated chloroplasts. Polyphenoloxidase in *Beta vulgaris*. *Plant Physiol.* 24, 1–9.
- Bates, L.S., Waldren, R.P., Teare, I.D., 1973. Rapid determination of free proline for water-stress studies. *Plant Soil* 39, 205–207.
- Biederman, L.A., Harpole, W.S., 2013. Biochar and its effects on plant productivity and nutrient cycling: a meta-analysis. *Glob. Change Biol. Bioenergy* 5, 202–214.
- Bradford, M.M., 1976. A rapid and sensitive method for the quantitation of microgram quantities of protein utilizing the principle of protein-dye binding. *Anal. Biochem.* 72, 248–254.
- Chintala, R., Mollinedo, J., Schumacher, T.E., Malo, D.D., Julson, J.L., 2014. Effect of biochar on chemical properties of acidic soil. *Arch. Agron. Soil Sci.* 60, 393–404.
- de Brito Neto, J.F., Beltrão, N.D.E.M., Rigon, J.P.G., Capuani, S., 2012. Effect of calcium and magnesium silicate on the growth of the castor oil plant subjected to salinity levels. *Semin. Cienc. Agrar.* 33, 2949–2958.
- Farhangi-Abri, S., Ghassemi-Golezani, K., 2018. How can salicylic acid and jasmonic acid mitigate salt toxicity in soybean plants? *Ecotoxicol. Environ. Saf.* 147, 1010–1016.
- Farhangi-Abri, S., Tavasolee, A., Ghassemi-Golezani, K., Torabian, S., Monirifar, H., Rahmani, H.A., 2020. Growth-promoting bacteria and natural regulators mitigate salt toxicity and improve rapeseed plant performance. *Protoplasma* 257, 1035–1047.
- Farhangi-Abri, S., Torabian, S., 2017. Antioxidant enzyme and osmotic adjustment changes in bean seedlings as affected by biochar under salt stress. *Ecotoxicol. Environ. Saf.* 137, 64–70.
- Farzadfar, S., Zarinkamar, F., Hojati, M., 2017. Magnesium and manganese affect photosynthesis, essential oil composition and phenolic compounds of *Tanacetum parthenium*. *Plant Physiol. Biochem.* 112, 207–217.
- Ghassemi-Golezani, K., Farhangi-Abri, S., 2018. Foliar sprays of salicylic acid and jasmonic acid stimulate H⁺-ATPase activity of tonoplast, nutrient uptake and salt tolerance of soybean. *Ecotoxicol. Environ. Saf.* 166, 18–25.
- Ghassemi-Golezani, K., Farhangi-Abri, S., 2019. Biochar alleviates fluoride toxicity and oxidative stress in safflower (*Carthamus tinctorius* L.) seedlings. *Chemosphere* 223, 406–415.
- Ghassemi-Golezani, K., Hosseinzadeh-Mahootchi, A., Farhangi-Abri, S., 2020. Chlorophyll a fluorescence of safflower affected by salt stress and hormonal treatments. *SN Appl. Sci.* 2, 1–9.
- Grieve, C.M., Grattan, S.R., 1983. Rapid assay for determination of water soluble quaternary ammonium compounds. *Plant Soil* 70, 303–307.
- Gunes, A., Inal, A., Taskin, M.B., Sahin, O., Kaya, E.C., Atakol, A.R.D.A., 2014. Effect of phosphorus-enriched biochar and poultry manure on growth and mineral composition of lettuce (*Lactuca sativa* L. cv.) grown in alkaline soil. *Soil Use Manag.* 30, 182–188.
- Guo, X., Ji, Q., Rizwan, M., Li, H., Li, D., Chen, G., 2020. Effects of biochar and foliar application of selenium on the uptake and subcellular distribution of chromium in *Ipomoea aquatica* in chromium-polluted soils. *Ecotoxicol. Environ. Saf.* 206, 111184.
- Hendershot, W.H., Lalonde, H., Duquette, M., 1993. Ion exchange and exchangeable cations. *Soil Sampl. Methods Anal.* 19, 167–176.
- Hussain, M.I., Lyra, D.A., Farooq, M., Nikoloudakis, N., Khalid, N., 2016. Salt and drought stresses in safflower: a review. *Agron. Sustain. Dev.* 36, 4.
- Javadi, T., Farooq, M.A., Akhtar, J., Saqib, Z.A., Anwar-ul-Haq, M., 2019. Silicon nutrition improves growth of salt-stressed wheat by modulating flows and partitioning of Na⁺, Cl⁻ and mineral ions. *Plant Physiol. Biochem.* 141, 291–299.
- Jones Jr., J.B. (Ed.), 1999. *Soil Analysis Handbook of Reference Methods*. CRC Press.
- Joseph, S., Anawar, H.M., Storer, P., Blackwell, P., Chee, C.H.I.A., Yun, L.I.N., Munroe, P., Donne, S., Horvat, J., Jianli, W.A.N.G., Solaiman, Z.M., 2015. Effects of enriched biochars containing magnetic iron nanoparticles on mycorrhizal colonisation, plant growth, nutrient uptake and soil quality improvement. *Pedosphere* 25, 749–760.
- Kochert, G., Hellebust, J.A., Craigie, J.S., 1978. *Handbook of Phycological Methods*. Cambridge University Press, London, pp. 455–456.
- Lateef, A., Nazir, R., Jamil, N., Alam, S., Shah, R., Khan, M.N., Saleem, M., 2019. Synthesis and characterization of environmental friendly corn cob biochar based nano-composite – a potential slow release nano-fertilizer for sustainable agriculture. *Environ. Nanotechnol. Monit. Manag.* 11, 100212.
- Machado, R.M.A., Serralheiro, R.P., 2017. Soil salinity: effect on vegetable crop growth. Management practices to prevent and mitigate soil salinization. *Horticulturae* 3, 30.
- Rahman, A., Hossain, M.S., Mahmud, J.A., Nahar, K., Hasanuzzaman, M., Fujita, M., 2016. Manganese-induced salt stress tolerance in rice seedlings: regulation of ion homeostasis, antioxidant defense and glyoxalase systems. *Physiol. Mol. Biol. Plants* 22, 291–306.
- Rivelli, A.R., De Maria, S., Pizzi, S., Gherbin, P., 2010. Growth and physiological response of hydroponically-grown sunflower as affected by salinity and magnesium levels. *J. Plant Nutr.* 33, 1307–1323.
- Rostamian, R., Heidarpour, M., Mousavi, S.F., Afyuni, M., 2015. Characterization and sodium sorption capacity of biochar and activated carbon prepared from rice husk. *J. Agr. Sci. Tech.* 17, 1057–1069.
- Shahi, S., Srivastava, M., 2018. Influence of foliar application of manganese on growth, pigment content, and nitrate reductase activity of *Vigna radiata* (L.) R. Wilczek under salinity. *J. Plant Nutr.* 41, 1397–1404.
- Sharma, A., Shahzad, B., Kumar, V., Kohli, S.K., Sidhu, G.P.S., Bali, A.S., Handa, N., Kapoor, D., Bhardwaj, R., Zheng, B., 2019. Phytohormones regulate accumulation of osmolytes under abiotic stress. *Biomolecules* 9, 285.
- Shen, Z., Zhang, J., Hou, D., Tsang, D.C., Ok, Y.S., Alessi, D.S., 2019. Synthesis of MgO-coated corn cob biochar and its application in lead stabilization in a soil washing residue. *Environ. Int.* 122, 357–362.
- Siddiqui, M.H., Al-Wahaibi, M.H., Firoz, M., Al-Khaishany, M.Y., 2015. Role of nanoparticles in plants. *Nanotechnology and Plant Sciences*. Springer, Cham, pp. 19–35.
- Silber, A., Levkovitch, I., Graber, E.R., 2010. pH-dependent mineral release and surface properties of corn straw biochar: agronomic implications. *Environ. Sci. Technol.* 44, 9318–9323.
- Silva, E.D.B., Ferreira, E.A., Pereira, G.A.M., Silva, D.V., Oliveira, A.J.M., 2017. Peanut plant nutrient absorption and growth. *Rev. Caatinga* 30, 653–661.
- Song, Z., Lian, F., Yu, Z., Zhu, L., Xing, B., Qiu, W., 2014. Synthesis and characterization of a novel MnOx-loaded biochar and its adsorption properties for Cu²⁺ in aqueous solution. *Chem. Eng. J.* 242, 36–42.
- Tan, X.F., Liu, Y.G., Gu, Y.L., Xu, Y., Zeng, G.M., Hu, X.J., Liu, S.B., Wang, X., Liu, S.M., Li, J., 2016. Biochar-based nano-composites for the decontamination of wastewater: a review. *Bioresour. Technol.* 212, 318–333.
- Torabian, S., Farhangi-Abri, S., Rathjen, J., 2018. Biochar and lignite affect H⁺-ATPase and H⁺-PPase activities in root tonoplast and nutrient contents of mung bean under salt stress. *Plant Physiol. Biochem.* 129, 141–149.
- Tränkner, M., Jaghdani, S.J., 2019. Minimum magnesium concentrations for photosynthetic efficiency in wheat and sunflower seedlings. *Plant Physiol. Biochem.* 144, 234–243.
- Wang, D., Jiang, P., Zhang, H., Yuan, W., 2020. Biochar production and applications in agro and forestry systems: a review. *Sci. Total Environ.* 723, 137775.

- Wen, P., Wu, Z., Han, Y., Cravotto, G., Wang, J., Ye, B.C., 2017. Microwave-assisted synthesis of a novel biochar-based slow-release nitrogen fertilizer with enhanced water-retention capacity. *ACS Sustain. Chem. Eng.* 5, 7374–7382.
- White, A.J., 2010. Development of an activated carbon from anaerobic digestion by-product to remove hydrogen sulfide from biogas. University of Toronto, Canada.
- Yao, Y., Gao, B., Chen, J., Zhang, M., Inyang, M., Li, Y., Alva, A., Yang, L., 2013. Engineered carbon (biochar) prepared by direct pyrolysis of Mg-accumulated tomato tissues: characterization and phosphate removal potential. *Bioresour. Technol.* 138, 8–13.
- Ye, Y., Cota-Ruiz, K., Hernandez-Viezcas, J.A., Valdes, C., Medina-Velo, I.A., Turley, R.S., Peralta-Videa, J.R., Gardea-Torresdey, J.L., 2020. Manganese nanoparticles control salinity-modulated molecular responses in *Capsicum annuum* L. through priming: a sustainable approach for agriculture. *ACS Sustain. Chem. Eng.* 8, 1427–1436.
- Yeilaghi, H., Arzani, A., Ghaderian, M., Fotovat, R., Feizi, M., Pourdad, S.S., 2012. Effect of salinity on seed oil content and fatty acid composition of safflower (*Carthamus tinctorius* L.) genotypes. *Food Chem.* 130, 618–625.
- Zama, E.F., Reid, B.J., Sun, G.X., Yuan, H.Y., Li, X.M., Zhu, Y.G., 2018. Silicon (Si) biochar for the mitigation of arsenic (As) bioaccumulation in spinach (*Spinacia oleracea*) and improvement in the plant growth. *J. Clean. Prod.* 189, 386–395.
- Zhang, H., Voroney, R.P., Price, G.W., White, A.J., 2017. Sulfur-enriched biochar as a potential soil amendment and fertiliser. *Soil Res.* 55, 93–99.